04-data-modeling-and-db-population

04 — Data Modeling & Database Population

Production database: PostgreSQL on Railway (managed plugin) **Local/dev database:** SQLite (`backend/ldapa.db` , auto-created on first run) **Schema source:** [`database/schema.sql`]() — compatible with both engines **Initial data source:** `export_members33756-1-1773244920final.csv` (Brilliant Directories LDAPA member export, 3,617 rows, 136 columns)

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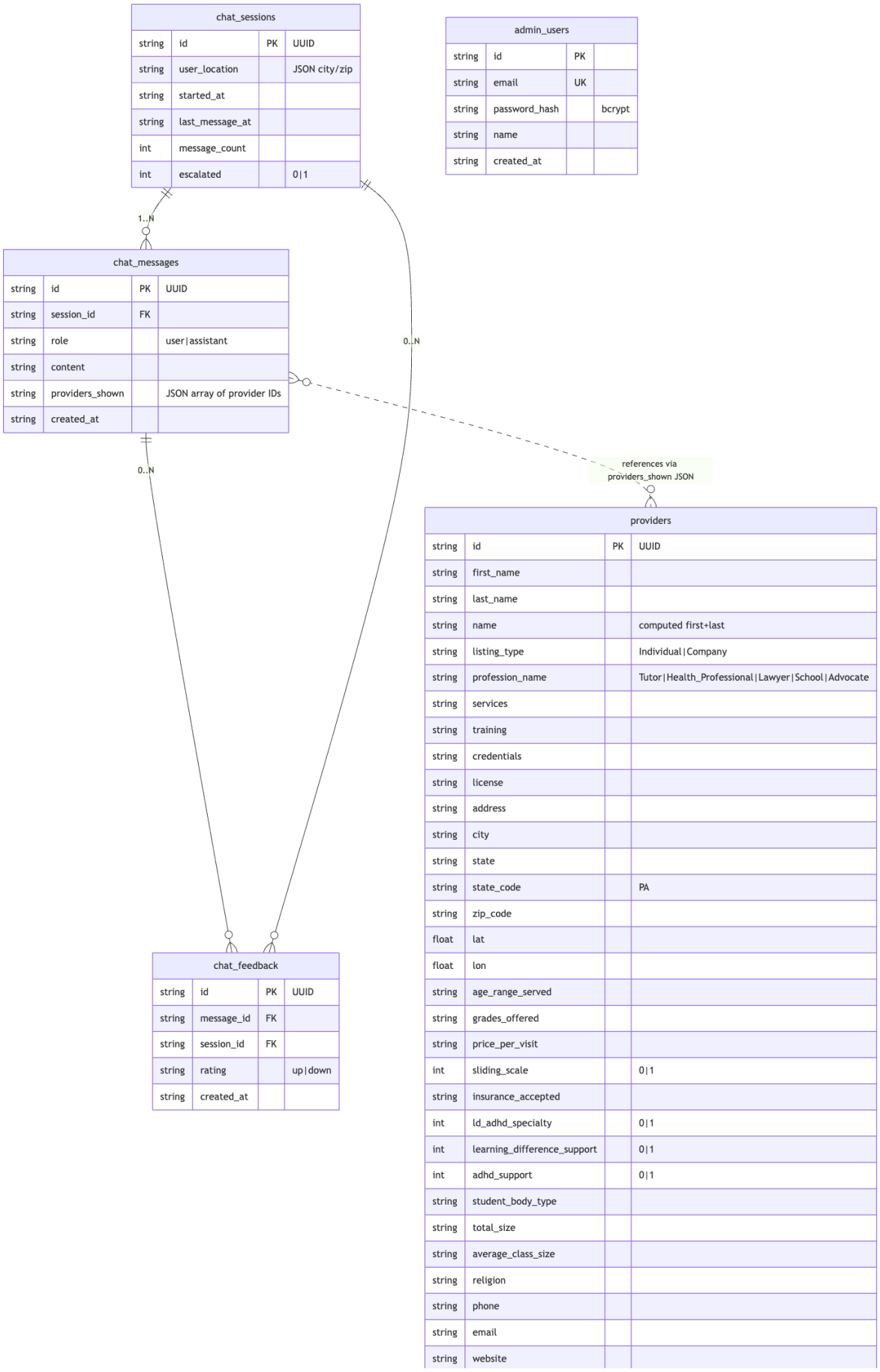
1. Entity-Relationship Diagram


chat_sessions
string id PK UUID
string user_location JSON city/zip
string started_at
string last_message_at
int message_count
int escalated 0|1
admin_users
string id PK
string email UK
string password_hash bcrypt
string name
string created_at
1..N
0..N
chat_messages
string id PK UUID
string session_id FK
string role user|assistant
string content
string providers_shown JSON array of provider IDs
string created_at
0..N
references via
providers_shown JSON
providers
string id PK UUID
string first_name
string last_name
string name computed first+last
string listing_type Individual|Company
string profession_name Tutor|Health_Professional|Lawyer|School|Advocate
string services
string training
string credentials
string license
string address
string city
string state
string state_code PA
string zip_code
float lat
float lon
string age_range_served
string grades_offered
string price_per_visit
int sliding_scale 0|1
string insurance_accepted
int ld_adhd_specialty 0|1
int learning_difference_support 0|1
int adhd_support 0|1
string student_body_type
string total_size
string average_class_size
string religion
string phone
string email
string website
chat_feedback
string id PK UUID
string message_id FK
string session_id FK
string rating up|down
string created_at

string	profile_url		
string	verification_status		unverified | verified | archived
string	last_verified_at		
string	staff_notes		
string	created_at		
string	updated_at		
int	is_deleted		0 | 1

ERD

Five tables:

Table	Rows (prod)	Purpose
`providers`	3,610	The LDA of PA directory — tutors, health professionals, lawyers, schools, advocates
`chat_sessions`	grows with usage	One row per conversation (anonymous)
`chat_messages`	grows with usage	One row per user or assistant turn
`chat_feedback`	grows with usage	Thumbs-up / thumbs-down events
`admin_users`	1	Staff login credentials (bcrypt-hashed)

Relationships:

- `chat_messages.session_id` → `chat_sessions.id` (ON DELETE CASCADE)
- `chat_feedback.session_id` → `chat_sessions.id` (ON DELETE CASCADE)
- `chat_feedback.message_id` → `chat_messages.id` (ON DELETE CASCADE)
- `chat_messages.providers_shown` is a JSON array of `providers.id` values — no FK constraint (many-to-many with soft coupling)
- `admin_users` is standalone (no FKs)

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2. Schema

2.1 `providers`

The core entity. Stores everything needed to display a provider card, rank it in search, and let staff edit it.

Column	Type	Notes
`id`	TEXT (PK)	UUID, generated on insert
`first_name`, `last_name`	TEXT	From CSV

Column	Type	Notes
`name`	TEXT NOT NULL	Computed “first last”
`listing_type`	TEXT	`Individual` or `Company`
`profession_name`	TEXT NOT NULL	One of `Tutor`, `Health_Professional`, `Lawyer`, `School`, `Advocate`
`services`	TEXT	Free text sub-services, e.g. `"Therapist,Doctor=>Psychiatrist"`
`training`	TEXT	Methodology, e.g. `"Wilson Language Training"`, `"Orton-Gillingham"`
`credentials`	TEXT	License numbers, e.g. `"Pennsylvania / SP023530"`
`license`	TEXT	License type, e.g. `"Psychologist"`
`address`	TEXT	Raw address from CSV `address1`
`city`	TEXT	Regex-parsed from address — see §4.2
`state`	TEXT DEFAULT `'PA'`	Full state name
`state_code`	TEXT DEFAULT `'PA'`	2-letter
`zip_code`	TEXT	5-digit
`lat`, `lon`	REAL	Used by geo search
`age_range_served`	TEXT	Free text, e.g. `"Adults, Teens"` or `"4-65"`
`grades_offered`	TEXT	School-only, e.g. `"Grades 1-8"`
`price_per_visit`	TEXT	Range, e.g. `"$150 - $250"`
`sliding_scale`	INTEGER (0/1)	Affordability flag
`insurance_accepted`	TEXT	Free text
`ld_adhd_specialty`	INTEGER (0/1)	LD/ADHD specialist flag
`learning_difference_support`	INTEGER (0/1)	Supports learning differences
`adhd_support`	INTEGER (0/1)	Supports ADHD
`student_body_type`	TEXT	School-only: `Co-Ed`, `All-boys`, `All-girls`
`total_size`	TEXT	School-only
`average_class_size`	TEXT	School-only

Column	Type	Notes
`religion`	TEXT	School-only
`phone`, `email`, `website`	TEXT	Contact info
`profile_url`	TEXT	Link to the Brilliant Directories listing
`verification_status`	TEXT NOT NULL DEFAULT `'unverified'`	`unverified` / `verified` / `archived`
`last_verified_at`	TEXT	ISO timestamp
`staff_notes`	TEXT	Admin-only notes
`created_at`, `updated_at`	TEXT NOT NULL	ISO timestamp, default `datetime('now')` / `CURRENT_TIMESTAMP`
`is_deleted`	INTEGER NOT NULL DEFAULT 0	Soft-delete flag

Indexes:

```
CREATE INDEX idx_providers_city       ON providers (city);
CREATE INDEX idx_providers_zip        ON providers (zip_code);
CREATE INDEX idx_providers_state      ON providers (state_code);
CREATE INDEX idx_providers_status     ON providers (verification_status);
CREATE INDEX idx_providers_profession ON providers (profession_name);
```

2.2 `chat_sessions`

Column	Type	Notes
`id`	TEXT (PK)	UUID
`user_location`	TEXT	JSON `{"city":"...","zip":"..."}`; populated only if the user mentioned location
`started_at`	TEXT NOT NULL	Default `datetime('now')` / `CURRENT_TIMESTAMP`
`last_message_at`	TEXT NOT NULL	Updated on every turn
`message_count`	INTEGER NOT NULL DEFAULT 0	Incremented by 2 per turn (user + assistant)
`escalated`	INTEGER NOT NULL DEFAULT 0	Set to 1 when the LLM detects a crisis

2.3 `chat_messages`

Column	Type	Notes
`id`	TEXT (PK)	UUID
`session_id`	TEXT NOT NULL	FK → `chat_sessions.id` ON DELETE CASCADE
`role`	TEXT NOT NULL	`user` or `assistant`
`content`	TEXT NOT NULL	The message body (max 10,000 chars)
`providers_shown`	TEXT DEFAULT `'[]'`	JSON array of provider IDs rendered with this message
`created_at`	TEXT NOT NULL	Default `datetime('now')`

Index: `idx_messages_session ON (session_id, created_at)` — used by the admin chat-transcript view.

2.4 `chat_feedback`

Column	Type	Notes
`id`	TEXT (PK)	UUID
`message_id`	TEXT NOT NULL	FK → `chat_messages.id` ON DELETE CASCADE
`session_id`	TEXT NOT NULL	FK → `chat_sessions.id` ON DELETE CASCADE
`rating`	TEXT NOT NULL	`up` or `down` (API accepts `positive` / `negative` and stores as `up` / `down`)
`created_at`	TEXT NOT NULL	Default `datetime('now')`

Index: `idx_feedback_session ON (session_id)`.

2.5 `admin_users`

Column	Type	Notes
`id`	TEXT (PK)	
`email`	TEXT NOT NULL UNIQUE	
`password_hash`	TEXT NOT NULL	bcrypt `$2b$12$...`
`name`	TEXT NOT NULL	

Column	Type	Notes
`created_at`	TEXT NOT NULL	

Default seed: - `id = 'admin1'` - `email = 'admin@ldapa.org'` - `password = 'admin123'` (hashed) - `name = 'LDA of PA Admin'`

This password must be changed after takeover. See **Doc 6 — Handoff**.

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3. Source Data: The Brilliant Directories CSV Export

All 3,610 production providers were imported from a single file: `export_members33756-1-1773244920final.csv` — an export of the LDA of PA directory maintained in the Brilliant Directories SaaS.

3.1 File characteristics

Property	Value
Filename	`export_members33756-1-1773244920final.csv`
Encoding	UTF-8 with BOM
Delimiter	Comma
Row count	3,617 data rows (3,610 accepted + 7 rejected)
Column count	136

The file is checked in at the repo root so that `init_db()` can auto-seed an empty database on first deploy.

3.2 Columns used (subset of 136)

Only the following CSV columns are read by the importer — the other ~100 columns (social media handles, appointment links, `cv_url`, `credentials_upload_url`, etc.) are ignored. See `backend/app/services/csv_importer.py` for the exact mapping.

CSV column	DB column	Transformation
`first_name`, `last_name`	`first_name`, `last_name`, `name`	Concatenated into `name`
`listing_type`	`listing_type`	Direct
`profession_name`	`profession_name`	Validated against 5 allowed values; rows with other values still import but log a warning
`services`	`services`	Direct
`training`	`training`	Direct

CSV column	DB column	Transformation
`credentials` / `Credentials`	`credentials`	First non-empty wins
`license`	`license`	Direct
`address1`	`address` , `city` (parsed)	Address stored as-is; city extracted via regex (see §4.2)
`state_ln`	`state`	Default `"Pennsylvania"`
`state_code`	`state_code`	Default `"PA"`
`zip_code`	`zip_code`	Direct
`lat` , `lon`	`lat` , `lon`	`_safe_float()` — returns `NULL` on `0` or parse failure
`agerangeserved`	`age_range_served`	Direct
`grades_offered`	`grades_offered`	Direct
`pricepervisit`	`price_per_visit`	Direct
`Slidingscaleoffered`	`sliding_scale`	`_parse_bool()` — handles `"0"` , `"1"` , `"$1.00"` , `"yes"/"no"`
`insurancesaccepted`	`insurance_accepted`	Direct
`LDADHDspeciality`	`ld_adhd_specialty`	`_parse_bool()`
`learning_difference_support`	`learning_difference_support`	`_parse_bool()`
`adhd_support`	`adhd_support`	`_parse_bool()`
`student_body_type` , `total_size` , `average_class_size` , `religion`	same	Direct (school-only)
`phone_number` / `PhoneNumber`	`phone`	First non-empty
`Email` / `email`	`email`	First non-empty; `@ldaofpadirectory.com` placeholders filtered out
`website` / `Website`	`website`	First non-empty
`profile_url`	`profile_url`	Direct

Rejected rows (fatal errors) produce entries in the `errors` list and are not inserted:

- Missing both `first_name` and `last_name`
- Missing `profession_name`

Non-fatal warnings (row still imported):

- Unknown `profession_name` (outside the 5 allowed values)
- Could not parse a city from `address1`

3.3 City parsing

The Brilliant Directories `city` column is nearly empty (2 rows populated), so cities are extracted from the `address1` free-text field with a two-pattern regex:

1. `"(?:.*,\s*)?([A-Z][a-zA-Z\s.\-']+?),\s*(?:PA|Pennsylvania)\b"` — matches "Pittsburgh" in `"123 Main St, Pittsburgh, PA 15213"`. The match is rejected if the candidate contains digits or is too short (street-name false positive).
2. `"^([A-Z][a-zA-Z\s.\-']+)$"` — matches the whole string when `address1` is bare text like `"East Stroudsburg"`.

If both fail, the importer tries the CSV `city` column, then `City`, then gives up and logs a warning. Net result: city fill rate is 34.3%, compared to 90.6% zip fill rate. Geo search compensates by using lat/lon directly when the user provides a ZIP.

3.4 Email placeholder filtering

Brilliant Directories auto-generates placeholder email addresses `@ldaofpadirectory.com` for members who didn't supply one. These are filtered out during import — the DB stores `NULL` instead so staff can see that contact data is genuinely missing.

3.5 Import triggers

The CSV can enter the database three ways:

Trigger	When it runs	Where
Auto-import on first boot	Whenever the `providers` table is empty at startup	`backend/app/database.py` → `_import_csv_seed()`
Manual production seed	One-time, on handoff of an empty production DB	`backend/seed_production.py` — connects directly via `DATABASE_URL`
Admin CSV upload	Any time, via the admin UI	`/import-export` page → `POST /api/admin/providers/import/preview` then `.../import/confirm`

The admin upload path does **not** deduplicate — the same provider can be inserted twice if the same CSV is uploaded twice. Staff should be aware of this when importing incremental updates.

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4. Database Analytics (Current Production Dataset)

The numbers below reflect the state of the database after the initial CSV import. Run the same queries after any further import to refresh.

4.1 Breakdown by profession

Profession	Count	% of directory
Tutor	1,529	42.4%
Health Professional	1,138	31.5%
Lawyer	792	21.9%
School	150	4.2%
Advocate	1	0.03%
Total	**3,610**	**100%**

> The single “Advocate” row is a data-quality anomaly — historically most advocates in the directory were classified under `Health_Professional`. Staff may want to reclassify some.

4.2 Listing type

Type	Count
Individual	3,288
Company	322

4.3 Top 10 cities

City	Count
Philadelphia	154
Pittsburgh	89
Bryn Mawr	34
Media	26
Lancaster	25
Allentown	24
West Chester	24
Doylestown	22
Newtown	22
Bala Cynwyd	17

390 distinct cities are represented in the data.

4.4 Top 10 ZIP codes

ZIP	Count	Area
19103	62	Center City Philadelphia
19083	49	Havertown
19063	45	Media
19102	34	Rittenhouse Square
19382	34	West Chester
18940	32	Newtown
18901	31	Doylestown
19380	31	West Chester
15219	30	Pittsburgh (Hill District)
19010	30	Bryn Mawr

4.5 Schools by student-body type (150 schools total)

Type	Count
Co-Ed	145
All-boys	4
All-girls	1

4.6 Training methodologies (top)

Methodology	Providers
Wilson Language Training	1,394
Graduates in Good Standing of the Children's Dyslexia Centers	110
Wilson Language Certified Trainer	3

(41.8% of providers have any `training` value populated.)

4.7 Fill rates by field

Field	Populated	Fill rate
`zip_code`	3,270	90.6%

Field	Populated	Fill rate
`lat` / `lon`	3,166	87.7%
`email`	1,551	43.0%
`training`	1,510	41.8%
`website`	1,403	38.9%
`phone`	1,372	38.0%
`city`	1,240	34.3%
`age_range_served`	1,072	29.7%
`credentials`	997	27.6%
`price_per_visit`	800	22.2%
`insurance_accepted`	770	21.3%

4.8 Specialty flags

Flag	True	%
`sliding_scale`	503	13.9%
`ld_adhd_specialty`	159	4.4%
`learning_difference_support`	34	0.9%
`adhd_support`	25	0.7%

4.9 Verification status

Status	Count
Unverified	3,610
Verified	0
Archived	0

All 3,610 providers are currently `unverified`. Going through the list and verifying providers is one of the main ongoing tasks for LDA of PA staff.

4.10 Database size

File	Size
`ldapa.db` (SQLite dev)	~2 MB

File	Size
PostgreSQL prod equivalent	~5–8 MB (estimate)

The database is tiny by modern standards. There is no scaling concern for the foreseeable future; the current tier on Railway has gigabytes of headroom.

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5. How the Database Changes Over Time

5.1 Data the system writes automatically

Table	Write path
`chat_sessions`	`POST /api/chat` — one row per new session
`chat_messages`	`POST /api/chat` — two rows per turn (user + assistant)
`chat_feedback`	`POST /api/feedback` — one row per thumbs-up/down

5.2 Data staff modify via the admin panel

Table	Via
`providers`	Provider CRUD pages, bulk verify/archive, CSV import
`admin_users`	(Not exposed — must be updated via SQL or a new endpoint)

5.3 Data nothing modifies

Chat history is **append-only from the app's perspective**. There is no UI to delete a session or redact a message. If a user asks for their data to be deleted (GDPR/CCPA-style), staff would need to run a SQL `DELETE` against `chat_sessions WHERE id = ?` — the CASCADE foreign keys will clean up `chat_messages` and `chat_feedback` automatically.

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6. Backup and Restore

6.1 Production (Railway Postgres)

Railway takes automatic daily snapshots of managed Postgres on paid plans. On the free plan, backups are manual — **see Doc 5 — Deployment** for how to take a manual `pg_dump` and where to store it.

6.2 Local (SQLite)

The local `ldapa.db` file is the entire database. To back up: `cp backend/ldapa.db backend/ldapa.db.bak`. To restore: copy it back.

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7. Running Your Own Analytics

From the Railway Postgres console (**Data** tab on the Postgres service), you can run arbitrary SQL. Examples:

Monthly chat volume:

```
SELECT date_trunc('month', started_at::timestamp) AS month, COUNT(*)
FROM chat_sessions
GROUP BY 1 ORDER BY 1 DESC;
```

Most-shown providers:

```
SELECT p.id, p.name, p.profession_name, COUNT(*) AS times_shown
FROM chat_messages m,
     jsonb_array_elements_text(m.providers_shown::jsonb) AS pid,
     providers p
WHERE p.id = pid
GROUP BY p.id, p.name, p.profession_name
ORDER BY times_shown DESC LIMIT 20;
```

Feedback rate by week:

```
SELECT date_trunc('week', created_at::timestamp) AS week,
       COUNT(*) FILTER (WHERE rating = 'up')   AS up,
       COUNT(*) FILTER (WHERE rating = 'down') AS down
FROM chat_feedback
GROUP BY 1 ORDER BY 1 DESC;
```

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8. Related Documents

- **01 — Technical Requirements and Design**
- **02 — System Architecture**
- **03 — APIs**
- **05 — Deployment**
- **06 — Handoff**